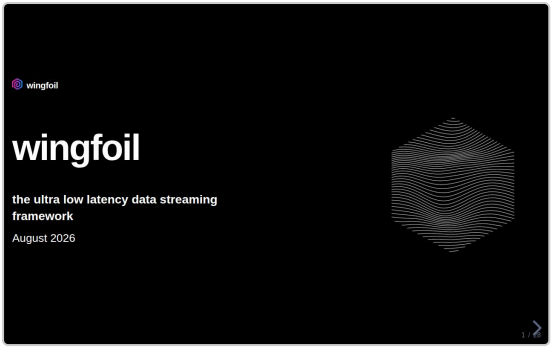
wingfoil — speaker notes

28 slides · printed fallback for the reveal.js speaker view (press S)



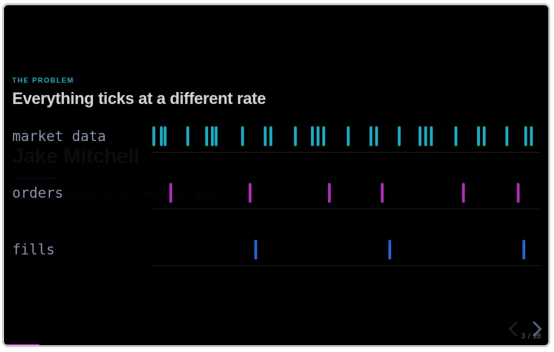
1 / 28

- Thanks for coming. Jake Mitchell. Open source project — **wingfoil**.
- Three parts:
- **1** — background: myself, the problem, and why you need wingfoil to solve it.
- **2** — **Nitro**. The feature we released last week. Why we wanted it. Why it meant rewriting the entire framework. The pattern we found, and what it delivered.
- **3** — what's next.

2 / 28

Jake Mitchell

- 25 years in financial services.
- Mostly as a quant at JPMorgan — building electronic trading systems for options: across interest rates, FX, precious metals.
- Now at B2C2, working for their crypto options desk.
- I learnt what the most effective patterns were for building these kinds of systems.
- I wanted to bring them into the open source world. And I wanted to use **Rust** to build it.

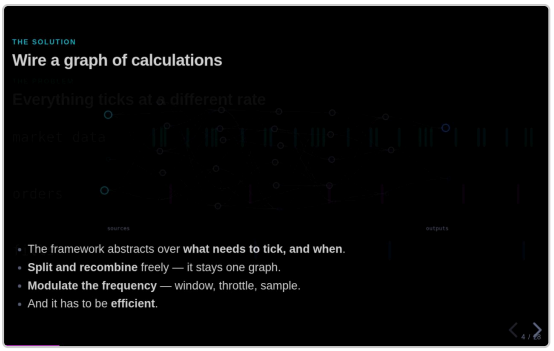


3 / 28

THE PROBLEM

Everything ticks at a different rate

- The problem.
- You're building something that reacts to information that is arriving at **different rates**.
- For example — market data ticking very frequently; orders and fills at a different frequency.
- This could be across tens of thousands of assets.
- And you need to react fast — at micro or even **nanosecond** time scales.



THE SOLUTION

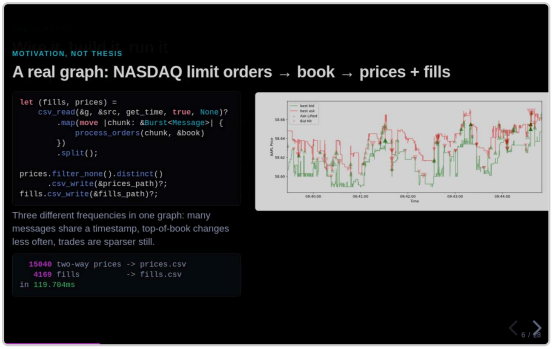
Wire a graph of calculations

- Wire a graph of calculations.
- Have a framework that **abstracts over what needs to tick at what time**.
- That allows you to split and recombine.
- And to modulate their frequency — for example window or throttle.
- And is **efficient**.

THE SURFACE

Wire it, build it, run it

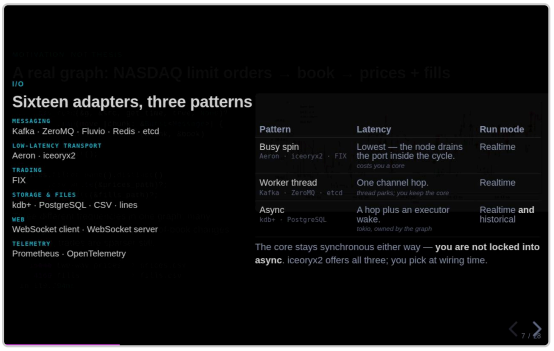
- This is what it looks like in practice.
- This example is a simple linear pipeline, with every node ticking on each engine cycle.



MOTIVATION, NOT THESIS

A real graph: NASDAQ limit orders → book → prices + fills

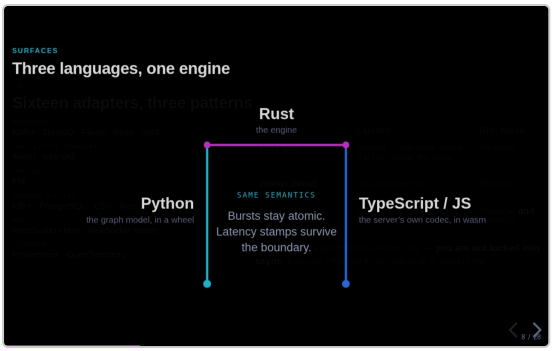
- Here is a more interesting example: a **matching engine**.
- Replay limit orders from file.
- Maintain an order book over time.
- Produce top-of-book prices and fills.



I/O

Sixteen adapters, three patterns

- We've built a bunch of I/O adapters — FIX, kdb+, iceoryx2, and more.
- They use these three I/O patterns:
- **Busy spin.**
- **Dedicated worker thread.**
- **Async.**



8 / 28

SURFACES

Three languages, one engine

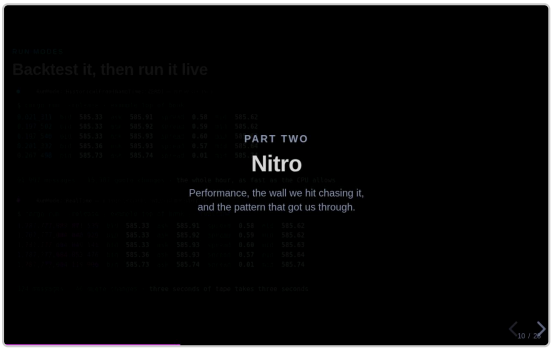
- **Rust** — the best language to work with.
- Also **Python** bindings, for those that have yet to make the jump.
- **TypeScript** integrations, to stream into the browser.

9 / 28

RUN MODES

Backtest it, then run it live

- Same wiring. Same graph. One line changes — the run mode.
- Top: the replay. It reads **no clock at all** — engine time is source-driven, so it goes as fast as the CPU walks the graph. An hour of AAPL in a fraction of a second.
- Bottom: same program, live. Rows arrive when the market gives them to you, and you can see it waiting in between.
- The quotes are **identical** across the two. Only the clock column differs — replay counts from zero, live is epoch wall-clock.
- That's the determinism: a backtest you can re-run and get the same answer.
- Second clock, separately: a wall-clock snap for latency stamping. Never branch business logic on it.
- And a live source **refuses a historical run at wiring time**, not at 3am — historical collects up front, so a live tail would deadlock.
- *[Live panel is slowed 12×, and says so — the five real gaps span a quarter of a second.]*

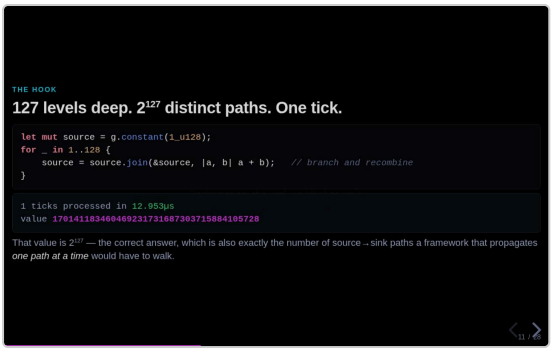


10 / 28

PART TWO

Nitro

- That's part one.
- Part two — Nitro.



THE HOOK

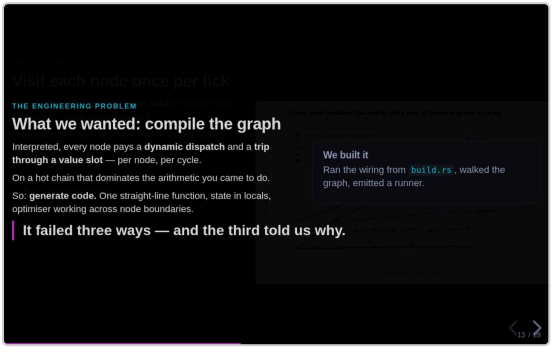
127 levels deep. 2127 distinct paths. One tick.

- 128 levels deep. Branches and recombines at every level.
- So distinct source→sink paths **double every level**.
- At the bottom: 2^{127} paths.
- That number on screen is the answer — **and** the path count.
- **Thirteen microseconds. One tick.**
- *[Slow down. Let it sit.]*

WHY IT STAYS FLAT

Visit each node once per tick

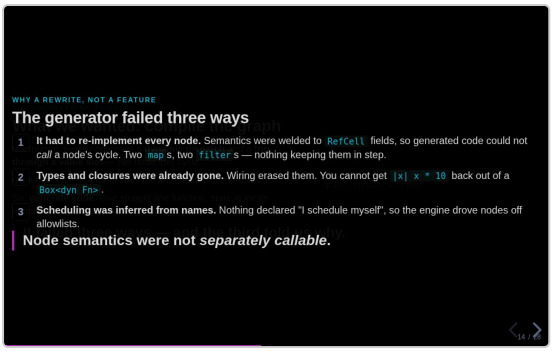
- The reason is boring, and that's the point.
- Every node runs once per tick, after everything it reads. However many paths lead in.
- Push one path at a time — reactive libraries, async streams — and you re-visit shared nodes per path. Cost doubles every level.
- Measured slopes: **2.01** and **1.94**. Ours: flat.
- *[Get to the caveats first: rxrust figure is HALVED from their raw number (their bench emits twice per sample). Tokio target is recursive Box::pin over a ready leaf — honest for per-path propagation, not the best async could do. **The claim is the slope, not the multiplier.**]*



THE ENGINEERING PROBLEM

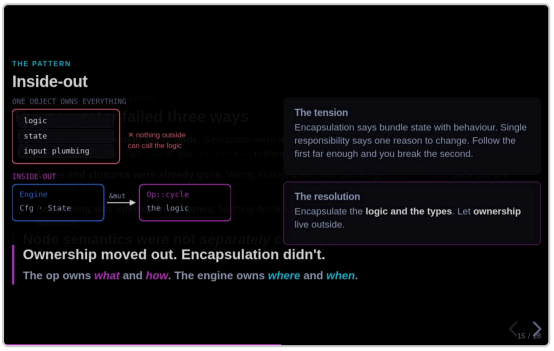
What we wanted: compile the graph

- Interpreted engine is fine, but there's overhead I don't love.
- Per node, per cycle: a **dynamic dispatch** and a **trip through a value slot**.
- On a hot chain that's bigger than the arithmetic you came to do.
- Obvious answer: **generate code.** One straight-line function, state in locals, optimiser working across node boundaries.
- Good plan. Every ingredient exists.
- So we built it. Wiring at build time, walk the graph, emit a runner.



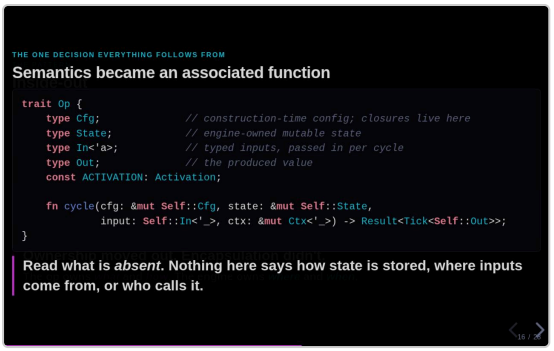
The generator failed three ways

- **It failed. Three ways.**
- **1** — had to re-implement every node. Semantics welded to RefCell fields, so generated code couldn't *call* a node's cycle. Emitter carried its own copy, as strings. Two maps, two filters. Nothing keeping them in step.
- **2** — types and closures already gone. Types came back as name strings we parsed. You cannot get a closure back out of a Box dyn Fn. Workaround: user retypes every closure in a side table, with no compiler check the two agree.
- *[That one usually gets a laugh. Let it.]*
- **3** — nothing declared how it wanted scheduling. Engine drove nodes off **allowlists of names**.
- We deleted it. *[Pause.]*
- **The problem was never the generator. Node semantics weren't separately callable.** No cleverness in an emitter fixes a design with nothing to emit a call *to*.



Inside-out

- The pattern that got us out.
- **The tension:** encapsulation says bundle state with behaviour. Single responsibility says one reason to change. Follow the first far enough and you break the second.
- Our node was computation *and* storage *and* plumbing. Three reasons to change, one object.
- **The resolution:** keep logic and types encapsulated. Move **ownership** out.
- *[Point at the arrow.]* The engine passes in state it *can't read*, to logic it *can't see inside*.
- **Ownership moved out. Encapsulation didn't.**
- **The op owns what and how. The engine owns where and when.**
- *["Isn't that just an ECS?" — yes, and reducers, gen_server, React hooks. Same family. Ours is the combination: reducer-shaped semantics + opaque associated types + a const for scheduling. That's what gets you two engines.]*

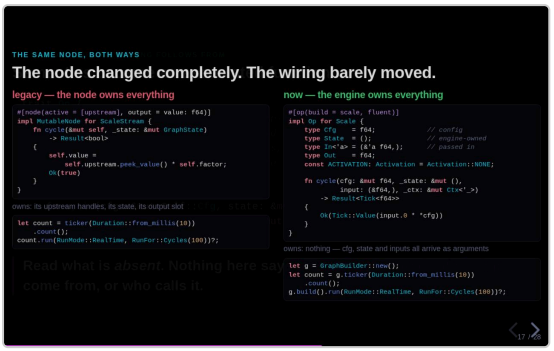


16 / 28

THE ONE DECISION EVERYTHING FOLLOWS FROM

Semantics became an associated function

- Four associated types and a function.
- Cfg (closures live here) · State (engine owns) · In (typed, per cycle) · Out. Plus a const for scheduling.
- **Notice what's missing.** Nothing says where state lives. Nothing says who calls it.
- It's a free function over explicit arguments.
- *[Return type, if asked: Tick has THREE states. Value ticks downstream, Quiet does nothing, Silent updates the value without ticking — what delay needs.]*

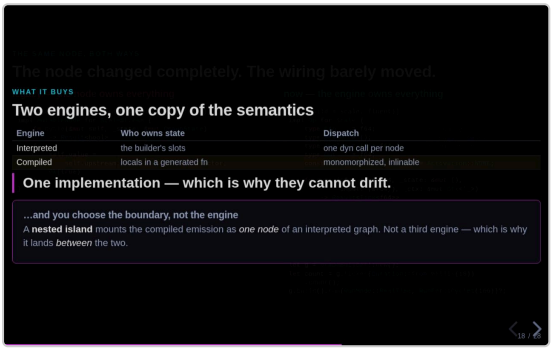


17 / 28

THE SAME NODE, BOTH WAYS

The node changed completely. The wiring barely moved.

- Same node, both ways.
- **Left:** &mut self, peeks a stored upstream handle, writes its own field. Owns its upstreams, its state, its output slot.
- **Right:** same node as an op. Owns **nothing** — everything arrives as an argument.
- Those two are unrecognisable.
- **Now the bottom row** — the wiring the user writes. Before and after.
- **One line different.** You hold a builder and make sources on it.
- We turned the engine inside out, and it cost anyone using it one line.



18 / 28

WHAT IT BUYS

Two engines, one copy of the semantics

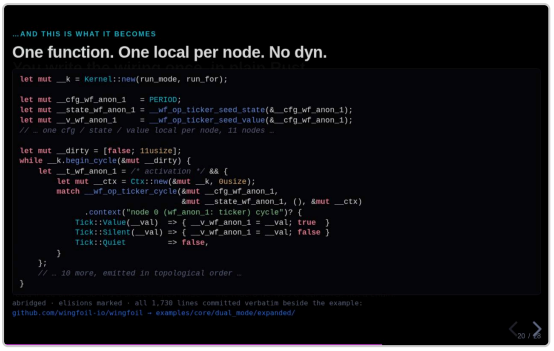
- Two engines out of one definition.
- Interpreted: state in the builder's slots, one dyn call per node.
- Compiled: state in locals of a generated fn, monomorphized, optimiser inlines through.
- **One implementation of what a node does.** They can't drift — nothing to drift from.
- Island: mount the compiled version as one node *inside* an interpreted graph. Hot core compiled, edges open.
- **Not a third engine — the seam between the two.** Which is why it lands between them on every benchmark.

19 / 28

THE FRONT DOOR

You write the wiring once, in plain Rust

- You write the wiring once. Plain Rust.
- Real function. Valid, reviewable, readable in a PR.
- Out the other side: interpreted, compiled, nested. **Same tokens.**
- `count` used three times → a shared node. This is a real DAG, not a chain.
- *[Constraint, if asked: wiring must be straight-line. Values and closure logic can be as procedural as you like — the TOPOLOGY can't depend on a runtime value, because compiled monomorphizes one local per node.]*

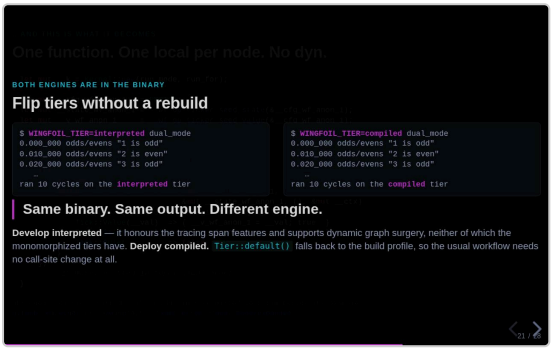


20 / 28

...AND THIS IS WHAT IT BECOMES

One function. One local per node. No dyn.

- What it turns into.
- Kernel on the stack. One local per node — cfg, state, value. Then a loop.
- *[Point at the call.]* **That's the same cycle function the interpreter calls.** Not a copy. A call.
- Wall number one, dead.
- Error context = the binding name *you* wrote, baked in as a static string. Costs nothing until something breaks.
- Whole expansion is committed. 1,730 lines. Don't take my word for it.

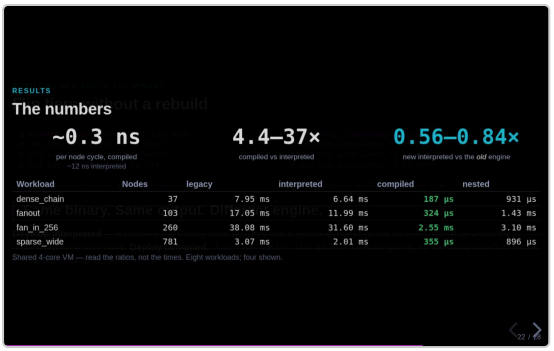


21 / 28

BOTH ENGINES ARE IN THE BINARY

Flip tiers without a rebuild

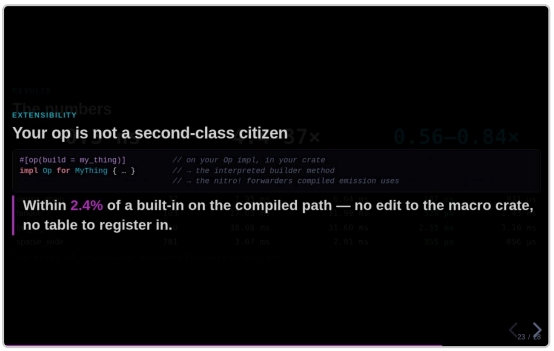
- Both engines are in the binary.
- Same binary, same output, different engine. **An environment variable.**
- Develop interpreted — proper tracing, surgery on a running graph. Neither of which compiled gives you.
- Deploy compiled.
- *[The two agreeing isn't a hope — a test runs the same graph through both and pins the outputs equal.]*



RESULTS

The numbers

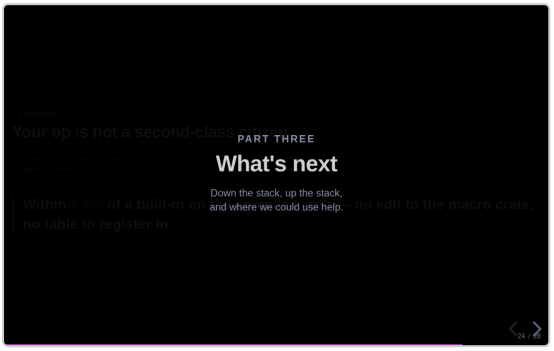
- **~0.3 ns** per node cycle compiled. **~12 ns** interpreted.
- Compiled **4.4× to 37×**, depending on graph shape.
- The one I care about most: **new interpreted beats the OLD engine. All eight workloads.**
- Because most graphs keep running interpreted — and a rewrite that trades everyday performance for a fast path nobody uses is a bad trade.
- *[Read the ratios, not the times. Shared cloud VM, measured back to back in one run.]*



EXTENSIBILITY

Your op is not a second-class citizen

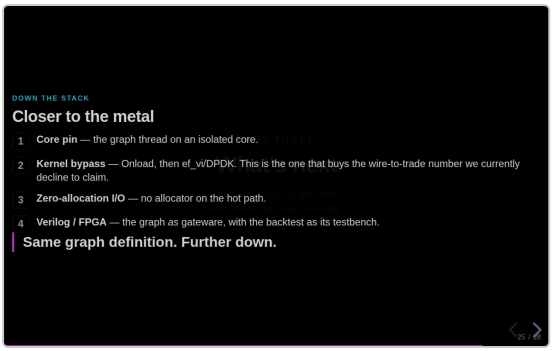
- The one I doubted most.
- Attribute on *your* op, in *your* crate → interpreted builder method + compiled forwarders.
- **Within 2.4% of a built-in** on the compiled path.
- No edit to the macro crate. No table to register in.
- There *is* no per-op table. Yours and ours take the same road.



PART THREE

What's next

- That's Nitro.
- Last bit — where this goes.

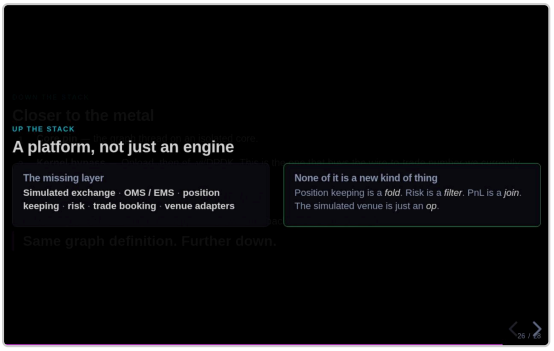


25 / 28

DOWN THE STACK

Closer to the metal

- **Core pin** — graph thread on an isolated core. Working implementation sits in an example; needs promoting into the runtime.
- **Kernel bypass** — Onload, then raw ef_vi/DPDK. Gets us a wire-to-trade number, which **I won't claim yet because I haven't measured it**.
- **Zero allocation** on the I/O path.
- **Verilog** — the graph as gateware, backtest as testbench. Exploratory. Don't oversell it.
- The point: **none of these change your wiring.** Same graph definition, further down. That's the payoff from part two.

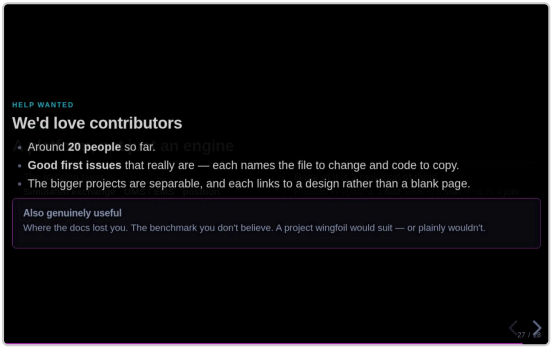


26 / 28

UP THE STACK

A platform, not just an engine

- Where most of the value is, for most people.
- **Be straight:** wingfoil is a compute engine. It is *not* a trading platform.
- Missing: simulated exchange, OMS, position keeping, risk, booking, more venue adapters.
- Simulator is the hard one — a naive fill-at-touch simulator lies to you, and you'll believe it.
- **The bet:** none of it is a new kind of thing. Position keeping is a *fold*. Risk is a *filter*. PnL is a *join*. The sim venue is just an *op* — RunMode picks.
- Ordinary graph vocabulary. Not a second framework bolted on top.

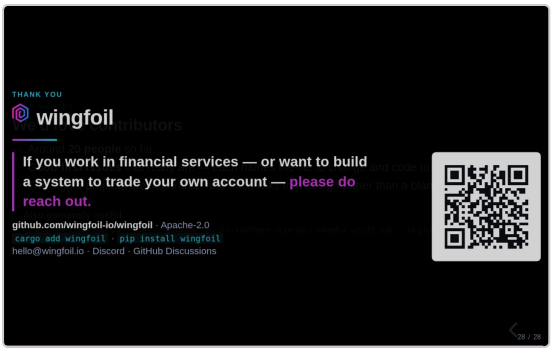


27 / 28

HELP WANTED

We'd love contributors

- The ask.
- ~20 contributors so far. I'd love more.
- **The good first issues are actually good first issues** — each names the file to change and code to copy. I know that label's been ruined elsewhere.
- Bigger projects are separable — one person each, and each links to a design not a blank page.
- Most useful isn't always code: where the docs lost you · which benchmark you don't believe · a project this would suit, or obviously wouldn't.



THANK YOU

wingfoil

- **If you work in financial services — or you're building something to trade your own account — please get in touch.** That's genuinely why I'm up here.
- Repo's on the QR. Thank you.
- *[Leave it up. Don't black the screen — let people scan.]*
- *[If it's quiet: the question I usually get is what happens when the graph shape isn't known until runtime. That's Project Lightning, and it's open.]*